PHYS 143 – Mechanics and Thermal Physics

The Second Law of Thermodynamics

Why do natural processes run in one direction only, and what limits how much of the heat from a hot source an engine can turn into work?

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This week

By the end of this week, you should be able to:

1. Distinguish reversible from irreversible processes, and calculate the entropy change of a system and its surroundings to decide whether a process is allowed.
2. Analyse a heat engine or a refrigerator with an energy-flow diagram, including from a pV cycle.
3. State the second law and use the Carnot cycle to find the maximum efficiency and to test a claimed device.
4. Analyse an unfamiliar engine, refrigerator or entropy problem, choosing the physics without being told.

The detailed outcomes are in the closing checklist. **Reading:** Young and Freedman, Ch. 20 (§20.1 to §20.7; §20.8 in the supplement).

Two contacts:

A (Tuesday): DIRECTION, then CONVERT
Why do processes run one way, what is entropy, and how does an engine use heat flow?

B (Wednesday): CONVERT, then LIMITS
How do engines and refrigerators move energy, what does the second law forbid, and what is the best engine?

Practice: check → example → independent → application

Not marked. Answer individually, then we review.

1. A gas goes round a clockwise cycle, absorbing $900\,\mathrm{J}$ in total and rejecting $600\,\mathrm{J}$ in total. Find the net work done by the gas and ΔU for the cycle.
2. How much heat warms $0.50\,\mathrm{kg}$ of water by $40\,\mathrm{K}$ ($c = 4186\,\mathrm{J/(kg\,K)}$)?
3. How much heat melts $0.10\,\mathrm{kg}$ of ice at $0^\circ\mathrm{C}$ ($L_f = 3.34 \times 10^5\,\mathrm{J/kg}$)?
4. Convert $27^\circ\mathrm{C}$ and $327^\circ\mathrm{C}$ to kelvin.
5. Evaluate $\displaystyle\int_{300}^{600} \frac{\mathrm{d}T}{T}$.
6. A diatomic gas ($\gamma = 1.40$) is compressed adiabatically to one-eighth of its volume. By what factor does T rise ($TV^{\gamma-1} = \text{constant}$)?

But what if... you want to know how you did? The next slide gives all six answers, each with its question restated.

Check: brief review

1. *A gas goes round a clockwise cycle, absorbing* $900\,\mathrm{J}$ *and rejecting* $600\,\mathrm{J}$ *in total. Find the net work and* ΔU*.*
   $W_{\text{net}} = Q_{\text{net}} = 900 - 600 = 300\,\mathrm{J}$; $\Delta U = 0$ for a cycle.
2. *How much heat warms* $0.50\,\mathrm{kg}$ *of water by* $40\,\mathrm{K}$ *(*$c = 4186\,\mathrm{J/(kg\,K)}$*)?*
   $Q = mc\Delta T = (0.50)(4186)(40) = 8.4 \times 10^4\,\mathrm{J}$.
3. *How much heat melts* $0.10\,\mathrm{kg}$ *of ice at* $0^\circ\mathrm{C}$ *(*$L_f = 3.34 \times 10^5\,\mathrm{J/kg}$*)?*
   $Q = mL_f = (0.10)(3.34 \times 10^5) = 3.34 \times 10^4\,\mathrm{J}$.
4. *Convert* $27^\circ\mathrm{C}$ *and* $327^\circ\mathrm{C}$ *to kelvin.*
   $300\,\mathrm{K}$ and $600\,\mathrm{K}$.
5. *Evaluate* $\int_{300}^{600} \mathrm{d}T/T$*.*
   $\ln(600/300) = \ln 2 = 0.693$.
6. *A diatomic gas (*$\gamma = 1.40$*) is compressed adiabatically to one-eighth of its volume. By what factor does* T *rise (*$TV^{\gamma-1} = \text{constant}$*)?*
   $T_2/T_1 = 8^{0.40} = 2.30$.

The cycle ($\Delta U = 0$), the ledger and the adiabatic relation (Lecture 11) and $Q = mc\Delta T$ and latent heat (Lectures 9 and 10) are all reused. Lecture 11 named the thermal efficiency and left it for this lecture.

But what if… these tools all handle energy and heat, yet none of them says which way a process may run? That is the first question of the lecture: **direction**.

Contact A: Direction, then Convert

Why do natural processes run only one way, what is entropy, and how does a heat engine use the flow of heat?

Poll

Four short films are run backwards. (A) A ball that bounces higher each time. (B) A hot cup of tea that gets hotter in a cool room. (C) Gas that gathers from a whole room into one corner. (D) A pendulum with no friction. **Which could really happen? Which break the first law?**

Predict first

Sketch each final state. For each film, is energy conserved? Is the film believable? Compare your two answers.

Only (D) looks natural. Films (A), (B) and (C) conserve energy, *so the first law does not forbid them*. Something else gives processes a direction. That is the second law.

But what if... the first law cannot tell (D) from (A), (B) and (C)? We need a sharper test of which processes can be undone and which cannot: **reversible and irreversible processes**.

Rule: reversible and irreversible

A process is **reversible** if its direction can be reversed by an infinitesimal change in the conditions, and the system is always in, or very close to, thermodynamic equilibrium (a smooth path of equilibrium states on a pV diagram). All other processes are **irreversible**: they proceed spontaneously in one direction but not the other.

Causes of irreversibility

- friction and viscosity (work becomes heat);
- heat flow across a *finite* temperature difference;
- free expansion into a vacuum;
- mixing of different substances.

Common error

"If energy is conserved, the process is possible." The first law does not give a direction. Real processes are irreversible; reversible ones are an idealisation that real processes approach.

But what if... the rule is still abstract? Test it on a block of ice in a metal box.

Ice in a metal box: two cases

(a) The box is at $70^\circ\mathrm{C}$

A block of ice at $0^\circ\mathrm{C}$ is placed in a metal box at $70^\circ\mathrm{C}$. The ice melts and the water warms: heat flows from the box into the ice and water, never the reverse.

(b) The box is at $0^\circ\mathrm{C}$

Ice at $0^\circ\mathrm{C}$ sits in a metal box at $0^\circ\mathrm{C}$. By raising or lowering the temperature of the box by an infinitesimal amount, we can make heat flow into the ice, so that it melts, or out of the water, so that it refreezes.

Predict first

Sketch the heat arrow in each case. In which case could the film be run backwards, and which is irreversible?

(a) is irreversible: heat flows across a finite temperature difference. **(b) is reversible:** the system stays in equilibrium with its surroundings throughout.

Everyday cases: bicycle tyres and brakes warm up as it stops (work becomes heat); an orderly system becomes disordered. Natural processes run from non-equilibrium to equilibrium states, from ordered to disordered; a refrigerator reverses this only with a supply of energy.

But what if... you must sort a new process with no film to watch? We need a habit: **Toolkit 1**.

- **TOOLKIT 1 – Reversible or irreversible?**

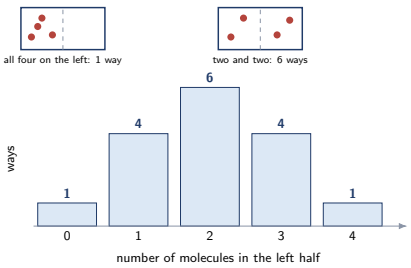
1. SKETCH the process and its surroundings, before and after.
2. RUN THE FILM BACKWARDS: would it look natural? If not, the process is irreversible.
3. LOOK FOR THE CAUSE: friction, a finite temperature difference, a sudden expansion, mixing.
4. COMPARE with the idealisation: quasi-static, equilibrium at every step, no friction.

WATCH OUT: energy conservation alone cannot decide; and "irreversible" does not mean "impossible", only "one direction".
CHECK: can both the system and the surroundings be restored without any net effect elsewhere?

But what if... the film test gives a verdict but no reason? Why is the one-way direction so overwhelmingly one-sided? We **count the ways**.

Why one direction: counting the ways

CORE



- Four molecules in a box: $2^4 = 16$ arrangements; only **1** puts all on the left, **6** give two and two.
- For 100 molecules the chance that all are on the left is $2^{-100} \approx 8 \times 10^{-31}$.
- Natural processes run from non-equilibrium to equilibrium, from ordered to disordered; a refrigerator forces the reverse only with an energy supply.

Physical meaning

Entropy is a quantitative measure of the randomness of a system: the spread-out state has overwhelmingly more microscopic arrangements, so it is overwhelmingly more likely and more random.

But what if... this count needs a name and a number? We define **entropy**, its change, and the **heat reservoir** that supplies heat at a fixed temperature.

Defining the vocabulary of direction

CORE

Quantity	What it is	What it measures	SI unit	Representation
Entropy S	a quantitative measure of the randomness of a system; a function of the state of the system	how random the system is (how many microscopic states it can be in)	J/K	a number for each state; a bar on a ledger
Entropy change ΔS	$S_2 - S_1 = \int \mathrm{d}Q/T$ along a reversible path from state 1 to state 2 (heat in is positive)	the increase or decrease in randomness, and so the direction a process may run	J/K	a signed number
Heat reservoir	a body that can give or take large amounts of heat at a constant temperature without appreciably changing its own temperature	the source or sink of heat	K	a labelled rectangle in the diagram

Interpret: entropy answers *in which direction can this process run?* (Reversible and irreversible processes were defined on the earlier Rule slide.)

But what if. . . knowing the words is not yet knowing the ideas? Test your intuition first, before any equation: a **concept check**.

Decide first

1. Can the entropy of a system decrease?
2. Can the total entropy of a system and its surroundings decrease?
3. Does heat ever flow from a cold body to a hot body by itself?
4. Is a melting ice cube an increase or a decrease of entropy?

Answers

1. **Yes.** Freezing water: heat leaves to the surroundings.
2. **No.** Counting system and surroundings together, entropy never decreases.
3. **No,** not without work from outside (a refrigerator).
4. **An increase:** the liquid has more arrangements.

But what if... yes or no is not enough, and we want a number? We need an equation for the entropy change: **the entropy rule**.

Rule: entropy change for a reversible isothermal process

$$\Delta S = S_2 - S_1 = \frac{Q}{T} \qquad [\mathrm{J/K}]$$

Q is the heat added *reversibly*: positive if heat flows into the system, negative if it flows out; T is the constant temperature in **kelvin**. In general $\mathrm{d}S = \mathrm{d}Q_{\mathrm{rev}}/T$.

Physical meaning

Heat flows in ($Q > 0$): the entropy of the system increases. Heat flows out ($Q < 0$): it decreases.

Common errors

(1) Using the heat of an irreversible process. (2) Using $^\circ\mathrm{C}$.

But what if... the real process is irreversible, so its own heat gives the wrong ΔS? We need to know what S depends on: **entropy is a state function**.

Rule: entropy is a function of state

$$\Delta S = S_2 - S_1 = \int_1^2 \frac{\mathrm{d}Q}{T}$$

along any reversible path from state 1 to state 2. The value depends only on the initial and final states, not on the path or the history, as for the internal energy U. For a gas, S is a function of the state variables, T, V and the amount of gas N.

Physical meaning

For an *irreversible* process, invent a path of reversible equilibrium processes between the same two states and calculate ΔS along it. It is not the actual path, but ΔS is the same.

Common error

Using the heat of the real, irreversible process in Q/T. In a free expansion $Q = 0$, yet $\Delta S > 0$ (Example 3).

But what if... each symbol in $\Delta S = Q/T$ still hides a choice (which heat, which temperature, which sign)? Read the equation piece by piece: **equation anatomy**.

EQUATION – entropy change

$$\Delta S = \frac{Q}{T}$$

Meaning. the entropy change $S_2 - S_1$ when heat Q is added reversibly at constant absolute temperature T.

Variables. Q heat added reversibly (positive into the system); T kelvin.

Units. $\mathrm{J/K}$.

Interpretation. heat in raises S; the same heat adds more entropy at low T.

Example. melting the ice: $+245\,\mathrm{J/K}$.

Predict first

Heat of $1000\,\mathrm{J}$ enters a body at $300\,\mathrm{K}$, and the same heat enters one at $600\,\mathrm{K}$. Predict which gains more entropy.

Answer: the colder body: 3.3 against $1.7\,\mathrm{J/K}$.

But what if... a problem has choices (constant T or not, which path, which sign)? A checklist: **Toolkit 2**.

TOOLKIT 2 – Entropy change

1. **IDENTIFY** the system and the initial and final states. Is T constant?
2. **CHOOSE A REVERSIBLE PATH** between the same states.
3. **SELECT:** constant T (phase change, reservoir): $\Delta S = Q/T$, with $Q = \pm mL$.
   Changing T: $\Delta S = mc\ln(T_2/T_1)$.
4. **PREDICT the sign** first: heat in at a given T gives $\Delta S > 0$; heat out gives $\Delta S < 0$.

WATCH OUT: kelvin only; do not use the heat of the irreversible process.
CHECK: units J/K; sign agrees with the prediction.

But what if... a checklist is only useful if it works on a real case? Melt some ice: **Example 1**.

Question

$1.00\,\mathrm{kg}$ of ice at $0^\circ\mathrm{C}$ melts reversibly to water at $0^\circ\mathrm{C}$ ($L_f = 3.34 \times 10^5\,\mathrm{J/kg}$). Find the entropy change of the ice and water.

Predict first

Sketch the heating curve: a flat section at $273\,\mathrm{K}$. Does the entropy of the system rise or fall? Is the number of ways larger in the liquid or in the solid?

Set up: constant $T = 273.15\,\mathrm{K}$; $Q = mL_f = (1.00)(3.34 \times 10^5) = 3.34 \times 10^5\,\mathrm{J}$ into the system.

Execute: $\Delta S = \dfrac{Q}{T} = \dfrac{3.34 \times 10^5}{273.15} = 1.22 \times 10^3\,\mathrm{J/K}$.

Evaluate: positive, as predicted: the liquid has more arrangements. Freezing the same water gives $-1.22 \times 10^3\,\mathrm{J/K}$ for the water. ✓

But what if... the water warms as heat flows in? Then Q/T has no single T: **entropy on heating**.

Rule: entropy change on heating

$$\Delta S = \int \frac{\mathrm{d}Q}{T} = \int_{T_1}^{T_2} \frac{mc\,\mathrm{d}T}{T} = mc \ln \frac{T_2}{T_1}$$

Constant c; T in kelvin; negative if the body cools.

Physical meaning

Heat the body in many small steps, each at almost constant T: add $\mathrm{d}S = mc\,\mathrm{d}T/T$ for each step. The same heat adds more entropy at a low temperature than at a high one.

Common error

Using $\ln(T_2/T_1)$ with Celsius values, or writing $\Delta S = mc\Delta T/T$ with one temperature. Use the logarithm of the *kelvin* ratio.

But what if... the result contains a logarithm of a kelvin ratio? Check each symbol and what to expect: **equation anatomy**.

EQUATION – entropy of heating

$$\Delta S = mc \ln \frac{T_2}{T_1}$$

Meaning. the sum of $\mathrm{d}Q/T$ as the temperature changes from T_1 to T_2.

Variables. m mass; c specific heat; T_1, T_2 in kelvin.

Units. $\mathrm{kg} \cdot \mathrm{J/(kg\,K)} = \mathrm{J/K}$.

Interpretation. positive on heating, negative on cooling.

Example. $0.50\,\mathrm{kg}$ water, 20 to $80^\circ\mathrm{C}$: $+389\,\mathrm{J/K}$.

Predict first

Predict the sign of the entropy change when a hot body cools, before using the equation.

Answer: negative: heat leaves, so $\Delta S < 0$.

But what if... the water was heated by something else, and that body changes entropy too? Ignoring it can hide an impossible result: count **system plus surroundings**.

Rule: the second law for entropy

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

No process is possible in which the total entropy decreases, when all systems that take part are included: $= 0$ for a reversible process; > 0 for an irreversible one; < 0 is **forbidden**. In a closed system the entropy can never decrease.

Physical meaning

The entropy of a *part* can fall (a freezer, a refrigerator) provided the entropy of the surroundings rises by at least as much. Unlike energy, entropy is not conserved.

Common error

"Entropy always increases." Only the *total* for system and surroundings (an isolated whole) cannot decrease.

But what if... you want to see what this says about each part and about the total? Read the **equation anatomy** and predict.

EQUATION – second law for entropy

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

Meaning. when all systems that take part are included, the total entropy remains constant or increases.

Variables. each part from $\Delta S = Q/T$ or $mc\ln(T_2/T_1)$.

Units. J/K.

Interpretation. > 0 irreversible, $= 0$ reversible, < 0 forbidden.

Example. $1500\,\text{J}$ from 400 to $300\,\text{K}$: $+1.25\,\text{J/K}$.

Predict first

Heat flows from a hot reservoir to a cold one. Predict the sign of each reservoir's entropy change and of the total.

Answer: hot $-$, cold $+$, total $+$ (Example 2).

But what if... several bodies and several signs must be kept straight? A bookkeeping habit: **Toolkit 3**.

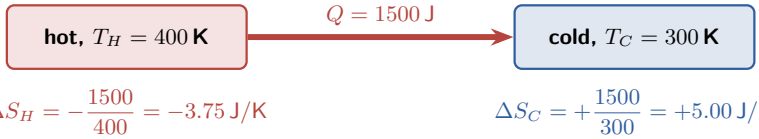
- **TOOLKIT 3 – Entropy bookkeeping**

1. LIST every body that gains or loses heat: system and each reservoir.
2. FOR EACH: $\Delta S_i = Q_i/T_i$ (reservoir, constant T) with Q_i signed for *that* body.
3. ADD: $\Delta S_{\text{total}} = \sum \Delta S_i$.
4. DECIDE: > 0 allowed and irreversible; $= 0$ allowed and reversible; < 0 **impossible**.

WATCH OUT: the hot body loses heat ($Q < 0$ for it); the cold body gains ($Q > 0$).
CHECK: the cold reservoir's gain exceeds the hot reservoir's loss, since the same $|Q|$ is divided by a smaller T.

But what if... the habit should reproduce the oldest observation of all, heat flowing from hot to cold? Check it in **Example 2**.

Example 2: heat flows from hot to cold



$$\Delta S_{\text{total}} = +1.25\,\mathbf{J/K} > 0$$

Question

A reservoir at $400\,\mathrm{K}$ passes $1500\,\mathrm{J}$ of heat to a reservoir at $300\,\mathrm{K}$. Find the entropy change of each reservoir and of the total. Then do the same for $1500\,\mathrm{J}$ flowing from the cold to the hot reservoir.

Predict first

Does the total entropy rise, fall or stay the same? Which of the two directions is allowed?

Example 2 restated

A reservoir at $400\,\mathrm{K}$ passes $1500\,\mathrm{J}$ of heat to one at $300\,\mathrm{K}$: find ΔS of each reservoir and of the total. Then repeat for $1500\,\mathrm{J}$ flowing from cold to hot.

Set up (Toolkit 3): each reservoir has constant T, so $\Delta S = Q/T$ with Q signed for *that* reservoir.

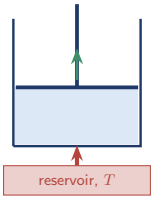
Forward: $\Delta S_{400} = \dfrac{-1500}{400} = -3.75\,\mathrm{J/K}$, $\Delta S_{300} = \dfrac{+1500}{300} = +5.00\,\mathrm{J/K}$, $\Delta S_{\text{total}} = +1.25\,\mathrm{J/K} > 0$: allowed.

Reverse: $\Delta S_{\text{total}} = +3.75 - 5.00 = -1.25\,\mathrm{J/K} < 0$: forbidden.

Evaluate: as predicted, heat flows from hot to cold by itself, never from cold to hot without help. ✓

But what if... heat flow is not the only irreversible process? A gas can expand into a vacuum with $Q = 0$ and nothing flows, so what does its entropy do? **Example 3**.

Example 3: expansion, reversible or free



reversible isothermal expansion



$Q = 0,\ W = 0$

free expansion (insulated)

$0.50\,\mathrm{mol}$ of an ideal gas at $300\,\mathrm{K}$ expands from $4.0\,\mathrm{L}$ to $12.0\,\mathrm{L}$, (i) reversibly and isothermally, taking heat from a reservoir at $300\,\mathrm{K}$ (left), or (ii) freely into a vacuum (right). Find ΔS of the gas and ΔS_{total} in each case.

Predict first

Free expansion has $Q = 0$. Is ΔS of the gas then zero? Which case is irreversible?

Why entropy rises: in an isothermal expansion of an ideal gas U depends only on T, so it stays constant; the gas does work, and its molecules move in a larger volume, so they are more random in position afterwards.

Example 3 restated

$0.50\,\text{mol}$ of an ideal gas at $300\,\text{K}$ expands from $4.0\,\text{L}$ to $12.0\,\text{L}$, (i) reversibly and isothermally from a reservoir at $300\,\text{K}$, (ii) freely into a vacuum. Find ΔS of the gas and ΔS_{total} in each case.

Set up: S is a state function, so ΔS_{gas} is the *same* for both. Use the reversible isothermal path: $Q = W = nRT\ln(V_2/V_1)$ (Lecture 11).

Gas: $\Delta S_{\text{gas}} = \dfrac{Q}{T} = nR\ln\dfrac{V_2}{V_1} = (0.50)(8.314)\ln 3 = 4.57\,\text{J/K} \quad (Q = 1.37\times10^3\,\text{J}).$

Reversible: the reservoir gives up $Q = 1.37\,\text{kJ}$ at $300\,\text{K}$: $\Delta S_{\text{surr}} = -4.57\,\text{J/K}$; $\Delta S_{\text{total}} = 0$.
Free expansion: $Q = 0$ for the surroundings: $\Delta S_{\text{surr}} = 0$; $\Delta S_{\text{total}} = +4.57\,\text{J/K} > 0$.

Evaluate: the gas ends in the same state, so its ΔS is equal; only the surroundings differ. The free expansion is irreversible, and its total entropy rises. ✓

But what if... the two irreversible processes so far gave different amounts of total entropy? Rank them in a **concept check**.

Choose, then explain

Rank the total entropy change, largest first. (i) A reversible isothermal expansion. (ii) $1500\,\mathrm{J}$ flows from $400\,\mathrm{K}$ to $300\,\mathrm{K}$. (iii) $1500\,\mathrm{J}$ flows from $400\,\mathrm{K}$ to $350\,\mathrm{K}$.

A. i, ii, iii B. ii, iii, i C. iii, ii, i D. ii, i, iii E. all equal

Predict first

Sketch the two heat flows. Which has the larger temperature difference? Which is closer to the reversible limit?

(ii) $+1.25$; (iii) $\dfrac{1500}{350} - \dfrac{1500}{400} = +0.54$; (i) 0. Ranking: **B**. The smaller the temperature difference, the closer to reversible: a small difference gives a small ΔS_{total}.

But what if... these calculations have several traps? Collect the habits that avoid them: **common mistakes**.

Common mistakes and the correct habit

Mistake	Instead
Thinking a process is possible because energy is conserved.	The first law gives no direction; test ΔS_{total}.
Using the heat of an irreversible process in $\Delta S = Q/T$.	Use a reversible path between the same states.
Using $^\circ\text{C}$ in an entropy formula.	Kelvin.
Saying entropy always increases.	Only the total of system and surroundings cannot decrease.
Forgetting the surroundings.	List every body that gains or loses heat.

But what if... the habits are in place? Take stock of the whole chain of direction and entropy.

The first question, answered

Why do natural processes run only one way, and what is entropy? Natural processes are irreversible: counting system and surroundings together, the total entropy increases, so they run only in the direction that raises it. Entropy is a quantitative measure of the randomness of a system.

Reversible or irreversible?

$\Downarrow$

Entropy: $\Delta S = Q/T,\ \ mc\ln(T_2/T_1)$; a state function

$\Downarrow$

Total: $\Delta S_{\text{total}} \geq 0$

You should be able to

- decide whether a process is reversible, and why;
- calculate ΔS for a phase change and for heating;
- find ΔS of system, surroundings and total;
- use ΔS_{total} to decide if a process is allowed.

So far

Entropy tells us which way a process may run, and heat flowing from a hot body to a cold one is the one-way flow we meet everywhere. The entropy rule decides which energy conversions are possible.

We now ask a new question

That one-way flow is not wasted: can a device use it to do work, and can work push heat the other way? How much work, and what does it cost?

Physical question	Tool
How much of the heat becomes work?	**energy-flow diagram, efficiency (Toolkit 4)**
How do we find it from a pV cycle?	**heat in and out leg by leg (Toolkit 5)**
What does it cost to move heat uphill?	**refrigerator, coefficient of performance (Toolkit 6)**

But what if... downhill heat can be put to work? That is a **heat engine**.

Sketch first

Heat flows from a hot reservoir to a cold one. Draw a device between them that produces work. Show every energy arrow and say where all the energy goes.

Predict first

Must all the heat taken from the hot reservoir become work? If the device returns to its starting state after each cycle, what does the first law say about the work?

The device must reject *some* heat to a cold reservoir. We now count the energy: the work out equals the heat in minus the heat rejected.

But what if... your sketch has arrows but no exact statements? Test your ideas on four short questions before any equation.

Concept check: engines before the equation

Decide first

1. Can an engine that runs in cycles turn all the heat it absorbs into work?
2. Can an engine have an efficiency above 1?
3. Is the work out of an engine equal to the heat in?
4. Does a refrigerator break the second law by moving heat from cold to hot?

Answers

1. **No.** Some heat must be rejected to a cold reservoir.
2. **No.** That would create energy.
3. **No.** The work is the heat absorbed minus the heat rejected.
4. **No.** It needs work, and the total entropy still rises.

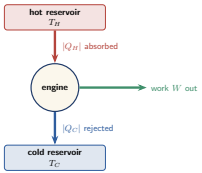
But what if... the answers use the words engine, efficiency and refrigerator? We must fix their exact meaning: the **vocabulary of engines**.

Defining the vocabulary of engines

Quantity	What it is	What it measures	SI unit	Representation
Heat engine	any device that transforms heat partly into work; its working substance goes round a cyclic process that returns it to its initial state	how heat is partly converted into work	—	hot reservoir T_H, engine, work W, cold reservoir T_C
Thermal efficiency e	the quotient of the net work W and the heat $\lvert Q_H\rvert$ absorbed from the hot reservoir: $e = W/\lvert Q_H\rvert$	the fraction of $\lvert Q_H\rvert$ converted to work: what you get divided by what you pay for	none	the width of the work arrow
Coefficient of performance K	for a refrigerator, the heat $\lvert Q_C\rvert$ removed from the cold place divided by the work input W: $K = \lvert Q_C\rvert/W$	how much heat is moved per joule of work; larger is better	none	the cold arrow over the work arrow

Interpret: thermal efficiency answers *how much of what I paid in heat do I get as work?*

But what if... these definitions must become an energy count: heat in, work out, heat rejected? That is the **energy balance of an engine**.



Rule: heat engine

$$W = |Q_H| - |Q_C|$$

$$e = \frac{W}{|Q_H|} = 1 - \frac{|Q_C|}{|Q_H|}$$

Heat is positive into the working substance, negative out: $Q_H > 0$, $Q_C < 0$. $|Q_H|$, $|Q_C|$ are **magnitudes** per cycle.

- Examples: automobile engines, jet engines, steam turbines in power plants. Every heat engine absorbs heat from a hotter source, does work and rejects heat at a lower temperature: the hot reservoir supplies $|Q_H|$, the cold reservoir (the sink) absorbs $|Q_C|$.
- The working substance returns to its start each cycle: $\Delta U = 0$, so $W = Q_{\text{net}}$ (Lecture 11).
- $0 < e < 1$; the energy in is shared between work and rejected heat.

But what if... the signs and ratios are easy to mix up in a problem? Read each symbol in the **equation anatomy**.

EQUATION – heat engine

$$W = |Q_H| - |Q_C|, \qquad e = \frac{W}{|Q_H|} = 1 - \frac{|Q_C|}{|Q_H|}$$

Meaning. the net work is the heat absorbed minus the heat rejected; the thermal efficiency is the fraction of $|Q_H|$ converted to work.

Variables. $|Q_H|$, $|Q_C|$ magnitudes of the heat in and out per cycle.

Units. J; e has no unit.

Interpretation. $\Delta U = 0$ over a cycle; $|Q_C| > 0$, so $e < 1$.

Example. $8000\,\mathrm{J}$ in, $5600\,\mathrm{J}$ out: $e = 0.30$.

Predict first

An engine absorbs $1000\,\mathrm{J}$ and rejects $700\,\mathrm{J}$. Predict its efficiency before calculating.

Answer: $W = 300\,\mathrm{J}$, $e = 0.30$.

But what if... a problem needs a drawing, a sign convention and the right ratio? A procedure: **Toolkit 4**.

TOOLKIT 4 – Energy flow in an engine

1. DRAW the hot and cold reservoirs, the engine and three arrows: $|Q_H|$ in, W out, $|Q_C|$ out.
2. USE MAGNITUDES: $W = |Q_H| - |Q_C|$ for each cycle (first law, $\Delta U = 0$).
3. EFFICIENCY: $e = W/|Q_H|$ (what you get over what you pay). For power, multiply per-cycle quantities by the cycle rate.
4. PREDICT: $|Q_C| > 0$ always, so $e < 1$; $W < |Q_H|$.
WATCH OUT: $e = W/|Q_C|$ or $|Q_C|/|Q_H|$ (wrong ratios); mixing signed and magnitude forms of Q.
CHECK: $|Q_H| = W + |Q_C|$, and $0 < e < 1$.

But what if... a procedure must be tested on numbers? Try it in **Example 4**.

Question

A petrol truck engine takes in $10\,000\,\mathrm{J}$ of heat and delivers $2000\,\mathrm{J}$ of mechanical work per cycle. Find (a) the efficiency, (b) the heat discarded in each cycle, (c) the power output at 25 cycles per second.

Predict first

Sketch the three arrows. Will e be more or less than $\frac{1}{2}$? Will the power be about 5, 50 or $500\,\mathrm{kW}$?

Execute: (a) $e = W/|Q_H| = 2000/10\,000 = 0.20$. (b) $|Q_C| = |Q_H| - W = 10\,000 - 2000 = 8000\,\mathrm{J}$. (c) $P = (2000)(25) = 5.0 \times 10^4\,\mathrm{W} = 50\,\mathrm{kW}$.
Evaluate: $e < \frac{1}{2}$ and about $50\,\mathrm{kW}$, as predicted; 0.20 means that 80% of the heat is rejected. ✓

But what if... a power plant is described by its power and efficiency, not by heats per cycle? Apply the same arrows to rates: **your turn**.

Independent problem

A power station delivers $600\,\mathrm{MW}$ of electrical power at an efficiency of 0.36. Find (a) the rate at which heat is supplied to it, (b) the rate at which heat is rejected to the river.

Predict first

Sketch the three arrows with widths in proportion. Is the rejected power more or less than the electrical power?

(a) heat supplied per second: $P_H = P/e = 600/0.36 = 1.67 \times 10^3\,\mathrm{MW}$. (b) heat rejected per second: $P_C = P_H - P = 1667 - 600 = 1.07 \times 10^3\,\mathrm{MW}$.

Evaluate: the rejected power ($1.07\,\mathrm{GW}$) exceeds the electrical power: a 36% plant rejects almost twice what it delivers. ✓

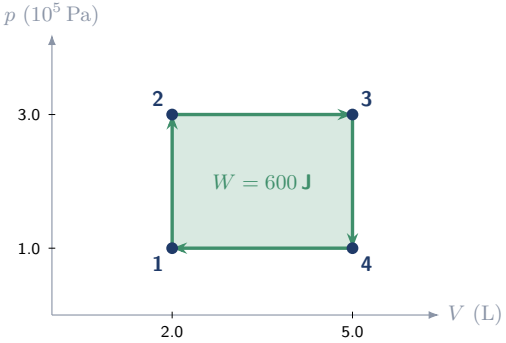
But what if... an engine is often given as a gas going round a pV cycle, with no heats stated? We must find the heat in and out leg by leg: **Toolkit 5**.

TOOLKIT 5 – Cycle analysis for efficiency

1. **LABEL** the states on the pV diagram; find each T from $pV = nRT$ (ratios are enough).
2. **LEDGER** (Lecture 11): for every leg find W, $\Delta U = nC_V\Delta T$ and $Q = \Delta U + W$.
3. **SORT** the legs: $Q > 0$ is heat *in*; $Q < 0$ is heat *out*. $|Q_H| = \sum Q_{>0}$, $|Q_C| = \sum |Q_{<0}|$.
4. **CLOSE:** $\sum \Delta U = 0$ and $W_{\text{net}} = |Q_H| - |Q_C|$; then $e = W_{\text{net}}/|Q_H|$.

WATCH OUT: an *isochoric* leg that raises T absorbs heat (no work, but $Q > 0$): it belongs in $|Q_H|$.
CHECK: net work equals the enclosed area.

But what if... a method for cycles needs a cycle with a hidden heat input? Try a rectangular one: **Example 5**.



A monatomic ideal gas ($C_V = \frac{3}{2}R$, $C_p = \frac{5}{2}R$) goes round the cycle $1 \to 2 \to 3 \to 4 \to 1$ shown: $1\,(1.0 \times 10^5\,\text{Pa},\ 2.0\,\text{L})$, $2\,(3.0 \times 10^5\,\text{Pa},\ 2.0\,\text{L})$, $3\,(3.0 \times 10^5\,\text{Pa},\ 5.0\,\text{L})$, $4\,(1.0 \times 10^5\,\text{Pa},\ 5.0\,\text{L})$. Find the net work, the heat absorbed per cycle and the efficiency.

Predict first

Which legs do you expect to take heat in? Is e more or less than 0.5?

Example 5 restated

A monatomic ideal gas ($C_V = \frac{3}{2}R$) goes round $1 \to 2 \to 3 \to 4 \to 1$, with $1\,(1.0 \times 10^5\,\text{Pa},\ 2.0\,\text{L})$, $2\,(3.0 \times 10^5\,\text{Pa},\ 2.0\,\text{L})$, $3\,(3.0 \times 10^5\,\text{Pa},\ 5.0\,\text{L})$, $4\,(1.0 \times 10^5\,\text{Pa},\ 5.0\,\text{L})$. Find the net work, the heat absorbed per cycle and the efficiency.

leg	W (J)	ΔU (J)	$Q = \Delta U + W$ (J)	rule used
$1 \to 2$ (isochoric, up)	0	$+600$	$+600$	$\Delta U = \frac{3}{2}V\Delta p$
$2 \to 3$ (isobaric, expand)	$+900$	$+1350$	$+2250$	$W = p\Delta V$
$3 \to 4$ (isochoric, down)	0	-1500	-1500	$\Delta U = \frac{3}{2}V\Delta p$
$4 \to 1$ (isobaric, compress)	-300	-450	-750	$W = p\Delta V$
total	$\mathbf{+600}$	$\mathbf{0}$	$\mathbf{+600}$	closes

But what if... the ledger gives every Q, but the question asks for the heat *absorbed* and the efficiency? We must sort the legs into heat in and heat out.

Example 5: solution (2) sort the legs

Asked: the heat absorbed per cycle and the efficiency, given the ledger of the previous slide: $Q = +600,\ +2250,\ -1500,\ -750\,\text{J}$ on legs $1 \to 2$, $2 \to 3$, $3 \to 4$, $4 \to 1$, and $W_{\text{net}} = +600\,\text{J}$.

Sort: heat *in* on legs $1 \to 2$ and $2 \to 3$: $|Q_H| = 600 + 2250 = 2850\,\text{J}$; heat out on $3 \to 4$ and $4 \to 1$: $|Q_C| = 1500 + 750 = 2250\,\text{J}$.

Result: $e = \dfrac{W}{|Q_H|} = \dfrac{600}{2850} = 0.21.$

Evaluate: $\sum \Delta U = 0$ and $W =$ enclosed area $= (2 \times 10^5)(3 \times 10^{-3}) = 600\,\text{J}$. ✓ As predicted, $e < 0.5$: the efficiency is low because the isochoric leg $1 \to 2$ costs heat but does no work.

But what if... the habits of a cycle ledger fail at a few predictable points? Collect them: **common mistakes in Contact A**.

Common mistakes and the correct habit

Mistake	Instead
Using $e = W/\lvert Q_C\rvert$ or mixing signed and magnitude forms.	$e = W/\lvert Q_H\rvert$, with magnitudes.
Leaving an isochoric heating leg out of $\lvert Q_H\rvert$.	Any leg with $Q > 0$ belongs in $\lvert Q_H\rvert$.
Adding W and Q with unlike signs in the ledger.	One row per leg, $Q = \Delta U + W$.

But what if... the habits are in place? Take stock of Contact A and look ahead.

The Contact A questions, answered

Processes run only in the direction that raises the total entropy of all the systems that take part; entropy measures randomness. A heat engine takes $|Q_H|$ from a hot reservoir, converts part to work W and rejects $|Q_C|$ to a cold reservoir, so $e = W/|Q_H| < 1$.

Reversible or irreversible?

$\Downarrow$

Entropy: $\Delta S = Q/T, \;\; mc\ln(T_2/T_1)$

$\Downarrow$

Total: $\Delta S_{\text{total}} \geq 0$

$\Downarrow$

Engine: $W = |Q_H| - |Q_C|, \;\; e = W/|Q_H|$;
pV cycle by legs

You should be able to

- decide whether a process is reversible, and why;
- calculate ΔS and find the total for system plus surroundings;
- use ΔS_{total} to decide if a process is allowed;
- draw the energy-flow diagram and find e from the heats or a pV cycle.

But what if... this cycle wastes heat on a leg that does no work? Contact B begins with the **Otto** cycle of a petrol engine.

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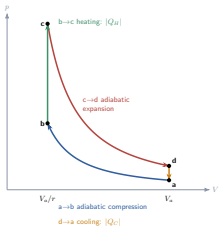
Contact B: Convert, then Limits

How do engines and refrigerators move energy, what does the second law forbid, and what is the best an engine can do?

Predict first

1. $1000\,\mathrm{J}$ of heat flows from a body at $500\,\mathrm{K}$ to a body at $250\,\mathrm{K}$. Find ΔS_{total} and say whether this is allowed. $\Delta S_{\text{total}} = -\dfrac{1000}{500} + \dfrac{1000}{250} = +2.0\,\mathrm{J/K} > 0$: allowed, and irreversible.
2. An engine takes in $800\,\mathrm{J}$ of heat per cycle and rejects $600\,\mathrm{J}$. Find W and e. $W = |Q_H| - |Q_C| = 800 - 600 = 200\,\mathrm{J}$, so $e = W/|Q_H| = 200/800 = 0.25$.
3. In a pV cycle, which legs count toward $|Q_H|$? Every leg on which $Q > 0$ (heat in), found from $Q = \Delta U + W$ leg by leg.

The internal-combustion engine: the Otto cycle



Idealised petrol engine, $r = V_a/V_b$ the compression ratio (largest volume over smallest):

1. **a**→**b** compression stroke (adiabatic);
2. **b**→**c** ignition (isochoric heating, $|Q_H|$);
3. **c**→**d** power stroke (adiabatic);
4. **d**→**a** exhaust (isochoric cooling, $|Q_C|$).

The real four-stroke engine: (1) *intake:* the piston moves down and the petrol–air mixture enters through the open intake valve; (2) *compression:* the intake valve closes and the mixture is compressed as the piston moves up; (3) *ignition:* the spark plug ignites the mixture; (4) *power:* the hot burned mixture expands and pushes the piston down; (5) *exhaust:* the exhaust valve opens and the piston moves up, expelling the burned gases, ready for the next intake. A typical r is 8 to 10. The Otto and Diesel cycles are irreversible: the arrows cannot be reversed without changing the temperatures of the reservoirs (the Diesel cycle is in the supplement).

Predict first

Will a larger r raise or lower the efficiency?

But what if... we can see the four strokes but not what fixes the efficiency? Work it out from the heats and the adiabats.

Heats: $|Q_H| = nC_V(T_c - T_b)$, $|Q_C| = nC_V(T_d - T_a)$, so $e = 1 - \dfrac{T_d - T_a}{T_c - T_b}$.

Adiabats ($TV^{\gamma-1}$ constant, Lecture 11): $T_b = T_a\, r^{\gamma-1}$, $T_d = T_c / r^{\gamma-1}$.

Combine: $T_d - T_a = \dfrac{T_c}{r^{\gamma-1}} - \dfrac{T_b}{r^{\gamma-1}} = \dfrac{T_c - T_b}{r^{\gamma-1}}$.

Rule: efficiency of the Otto cycle

$$e = 1 - \frac{1}{r^{\gamma-1}}$$

Ideal gas with constant C_V, reversible adiabats; it depends only on r and γ; a larger r raises e.

But what if... the result is compact, but what does each symbol say? Read the **equation anatomy**.

EQUATION – Otto cycle efficiency

$$e = 1 - \frac{1}{r^{\gamma-1}}, \qquad r = \frac{V_a}{V_b}$$

Meaning. the thermal efficiency of an idealised petrol engine.

Variables. r compression ratio; $\gamma = C_p/C_V$.

Units. none.

Interpretation. rises with r but stays below 1; independent of the heat supplied.

Example. $r = 8.0$, $\gamma = 1.40$: $e = 0.565$.

Predict first

Predict whether raising the compression ratio from 8 to 12 raises e by a little or by a lot.

Answer: by a little: $0.565 \to 0.630$ (see the concept check).

But what if... a formula should be tested on a full engine, with temperatures and heats? **Example 6**.

Question

$0.020\,\text{mol}$ of air ($C_V = \frac{5}{2}R$, $\gamma = 1.40$) is at $300\,\text{K}$ at the start of compression in an Otto cycle with $r = 8.0$. Ignition adds $|Q_H| = 500\,\text{J}$. Find the temperatures T_b, T_c, T_d, the heat rejected, the work and the efficiency.

Predict first

Sketch the cycle. Will e be about 0.3, 0.55 or 0.9? Which of T_b, T_c is the largest temperature?

Execute: $nC_V = (0.020)(20.79) = 0.416\,\text{J/K}$. $T_b = 300(8.0)^{0.40} = 689\,\text{K}$.
$T_c = T_b + |Q_H|/(nC_V) = 689 + 1203 = 1892\,\text{K}$. $T_d = T_c/8.0^{0.40} = 824\,\text{K}$.
$|Q_C| = nC_V(T_d - T_a) = (0.416)(524) = 218\,\text{J}$; $W = 500 - 218 = 282\,\text{J}$; $e = 282/500 = 0.56$.
Evaluate: formula $e = 1 - 8.0^{-0.40} = 0.565$. ✓ Real petrol engines reach about 0.25 to 0.35: friction, heat losses and non-adiabatic strokes.

But what if... a higher compression ratio looks like free efficiency? Test the prediction in a **concept check**.

Choose, then explain

Raising the compression ratio from 8 to 12 for $\gamma = 1.40$:

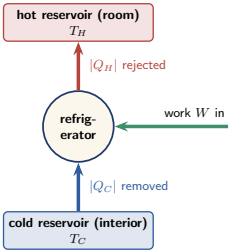
A. e falls B. e rises from 0.565 to 0.630 C. e rises to 0.90 D. e is unchanged E. e doubles

Predict first

Sketch e against r: rising steeply at first and then flattening? Does e reach 1 for any r?

$e = 1 - 12^{-0.40} = 1 - 0.370 = 0.630$: **B**. The curve flattens; $e < 1$ for every finite r. In a real petrol engine a high r ignites the fuel too early (knocking), which limits r to about 10 to 13.

But what if... an engine could be run backwards, with work supplied from outside? Then heat moves from cold to hot: the **refrigerator**.



Rule: refrigerator

$$|Q_H| = |Q_C| + W$$

$$K = \frac{|Q_C|}{W}$$

W: magnitude of the work input; K: coefficient of performance.

- A refrigerator takes heat from a cold place and gives it off to a warmer place; this needs a net work input W: it is a heat engine run in reverse.
- K can exceed 1. A good K is large.

But what if... the symbols and signs must be checked before use? Read the **equation anatomy**.

EQUATION – refrigerator

$$|Q_H| = |Q_C| + W, \qquad K = \frac{|Q_C|}{W}$$

Meaning. a refrigerator takes heat from a cold place and gives it off to a warmer place, which needs a net input of work.

Variables. $|Q_C|$ heat removed from the cold space; W work input; $|Q_H|$ heat released.

Units. J; K has no unit.

Interpretation. K can exceed 1; the hot side receives more than the cold side loses.

Example. $600\,\text{J}$ removed, $200\,\text{J}$ work: $K = 3.0$.

Predict first

A refrigerator moves $400\,\text{J}$ from inside using $100\,\text{J}$ of work. Predict the heat delivered to the kitchen.

Answer: $|Q_H| = 400 + 100 = 500\,\text{J}$.

But what if... the drawing, the energy balance and the ratio must all be right? A procedure: **Toolkit 6**.

- **TOOLKIT 6 – Refrigerators and heat pumps**

1. DRAW the same diagram as the engine with the arrows reversed: $|Q_C|$ in from the cold space, W in, $|Q_H|$ out to the hot space.
2. ENERGY: $|Q_H| = |Q_C| + W$ (first law for a cycle).
3. PERFORMANCE: refrigerator $K = |Q_C|/W$; heat pump (heating the hot space) $K_{\text{hp}} = |Q_H|/W$.
4. PREDICT: $|Q_H| > |Q_C|$: the hot space receives more heat than the cold space loses.
WATCH OUT: K is not an efficiency, and it is not limited to 1.
CHECK: $|Q_H| - |Q_C| = W > 0$.

But what if... a procedure must be tried on numbers? **Example 7**.

Question

In each cycle a refrigerator removes $600\,\mathrm{J}$ from its interior using $200\,\mathrm{J}$ of electrical work. Find K and the heat released into the kitchen.

Predict first

Sketch the arrows. Is the heat rejected to the kitchen more than, equal to, or less than $600\,\mathrm{J}$?

$$K = \frac{|Q_C|}{W} = \frac{600}{200} = 3.0. \qquad |Q_H| = |Q_C| + W = 600 + 200 = 800\,\mathrm{J}.$$

Evaluate: $800 > 600$, as predicted: the kitchen gains $800\,\mathrm{J}$ per cycle. ✓

But what if... a refrigerator moves heat out of its box, so can you cool a kitchen by leaving the door open? One last **concept check**.

Choose, then explain

A refrigerator is left open in a closed, insulated kitchen. The kitchen:

A. cools B. stays at the same temperature C. warms D. depends on K

Predict first

Sketch the kitchen as one system with electricity entering. What is the net energy flow across the walls?

C. The heat removed from the air is released back into the same room, plus the work: the net effect is $|Q_H| - |Q_C| = W > 0$. The kitchen warms by the electrical energy used. A refrigerator cools only if its hot side is outside the room.

But what if... the engine and refrigerator are now both in hand? Take stock, then ask what limits them.

So far

We can find e from the heats and from a pV cycle, and every engine we met had $e < 1$. The Carnot limit follows from the second law and sets the best efficiency of every power station and refrigerator.

We now ask a new question

Why can e never reach 1, and what is the best an engine can do between two given temperatures? Is the limit set by materials, or by nature?

Physical question	**Tool**
What exactly is forbidden?	**the three statements of the second law**
What is the best efficiency?	**the Carnot cycle, e_C and K_C (Toolkit 7)**
Is a claimed device possible?	**compare with e_C and ΔS_{total}**

But what if... an engine could take all its heat from one source and reject none? The next part begins with exactly that: a ship that runs on the heat of the ocean.

Sketch first

A ship's engine takes heat from the ocean at $290\,\mathrm{K}$ and turns *all* of it into work to drive the propeller. The ocean holds far more energy than the ship will ever use. Does this violate the first law? Why can it not be built?

Predict first

Sketch the energy-flow diagram. How many reservoirs does the device use? What is $|Q_C|$?

It does *not* violate the first law ($W = |Q_H|$ conserves energy). It uses one reservoir and has $|Q_C| = 0$, so $e = 1$. The second law forbids exactly this.

But what if... the ship obeys the first law yet cannot be built? We need the exact statement of what the second law forbids: **three statements**.

Rule: the second law of thermodynamics

Engine (Kelvin–Planck): It is impossible for any system to undergo a process in which it absorbs heat from a reservoir at a single temperature and converts the heat completely into mechanical work, with the system ending in the same state in which it began.

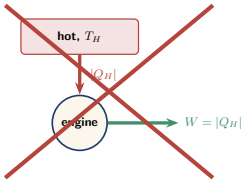
Refrigerator (Clausius): It is impossible for any process to have as its sole result the transfer of heat from a cooler to a hotter object.

Entropy: No process is possible in which the total entropy decreases, when all systems that take part in the process are included.

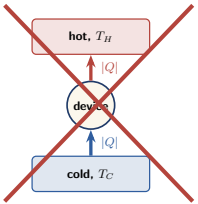
The three are equivalent: any device that violates one form can be used to make a device that violates the others. *The ship:* its engine takes heat from the ocean at $290\,\mathrm{K}$ and turns all of it into work. It does not violate the first law ($W = |Q_H|$ conserves energy), but it violates the engine statement: one reservoir, no heat rejected.

But what if... a statement about devices and a statement about entropy must agree if they are the same law? **Check** that each forbidden device lowers the total entropy.

The two forbidden devices



Kelvin–Planck: forbidden
(all heat into work)



Clausius: forbidden
(cold to hot, no work)

Kelvin–Planck: $\Delta S_{\text{total}} = -|Q_H|/T_H < 0.$

Clausius: $\Delta S_{\text{total}} = \dfrac{|Q|}{T_H} - \dfrac{|Q|}{T_C} < 0$ for $T_H > T_C$.

Explain. Each device would make the total entropy fall. Turning all the heat into work removes entropy $|Q_H|/T_H$ from the hot reservoir and adds none anywhere else. Moving heat $|Q|$ from cold to hot with no other effect takes less entropy from the cold body ($|Q|/T_C$ is the larger) than it adds to the hot body ($|Q|/T_H$), because $T_H > T_C$; the net change is negative.

But what if... all-heat-to-work is forbidden, so how much can an engine convert? We need names for the best case: the **vocabulary of limits**.

Defining the vocabulary of limits

Quantity	What it is	What it measures	SI unit	Representation
Carnot cycle	a cycle of two reversible isothermal and two reversible adiabatic processes; heat is exchanged only on the isotherms	the idealised cycle of greatest efficiency between two reservoirs	—	a closed loop of two isotherms and two adiabats on a pV diagram
Carnot efficiency e_C	the thermal efficiency of an engine that uses the Carnot cycle, $e_C = 1 - T_C/T_H$ (T in kelvin; written e_{Carnot} in the textbook)	the upper limit to the efficiency of any engine between the same two reservoirs	none	a limit line on a graph of e
Carnot coefficient of performance K_C	the coefficient of performance of the Carnot cycle run backward as a refrigerator, $K_C = T_C/(T_H - T_C)$	the greatest K any refrigerator can have between the same two temperatures	none	a limit line on a graph of K

Interpret: the Carnot efficiency answers *what is the best that nature allows between these two temperatures?*

But what if... these are only names for the best case? A **theorem** says why no engine can beat it.

Rule: Carnot's theorem

No engine can be more efficient than a **Carnot engine** operating between the same two temperatures. All Carnot engines operating between the same two temperatures have the **same** efficiency, irrespective of the working substance.

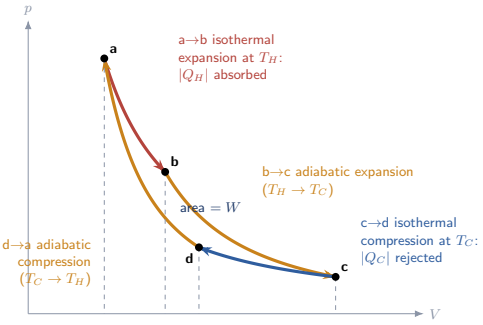
Physical meaning

The limit is set by nature, not by materials or design. It depends only on the two temperatures. This is a further equivalent form of the second law.

Common error

"A better working substance gives a better limit." It does not: every Carnot engine between T_H and T_C has the same e.

But what if... the theorem names the best engine but gives neither its cycle nor its efficiency? Build it from reversible steps: the **Carnot cycle**.



Two reversible isothermal and two reversible adiabatic processes (ideal gas shown). Every heat transfer is isothermal, because heat flow through a finite temperature difference is irreversible.

Predict first

Which curves are steeper, the isotherms or the adiabats? At which stages is heat exchanged?

The adiabats are steeper. Heat flows only on the isotherms (at T_H in, at T_C out); on the adiabats $Q = 0$ and the temperature changes.

But what if... we can see the four steps but not yet what they give? Find the heats on the two isotherms.

Isotherms (Lecture 11, $Q = W = nRT\ln(V_f/V_i)$): $|Q_H| = nRT_H \ln\dfrac{V_b}{V_a}$,

$|Q_C| = nRT_C \ln\dfrac{V_c}{V_d}$.

Adiabats ($TV^{\gamma-1}$ constant): $T_H V_b^{\gamma-1} = T_C V_c^{\gamma-1}$ and $T_H V_a^{\gamma-1} = T_C V_d^{\gamma-1}$.
Dividing: $V_b/V_a = V_c/V_d$, so the two logarithms are *equal*.

Rule: heat transfer in a Carnot engine

$$\frac{|Q_C|}{|Q_H|} = \frac{T_C}{T_H}$$

T in **kelvin**. Shown here for an ideal gas; it holds for any Carnot engine.

But what if... the ratio of the heats is fixed by the two temperatures alone? Then the efficiency and the refrigerator coefficient follow at once.

Rule: Carnot efficiency and refrigerator

$$e_C = 1 - \frac{|Q_C|}{|Q_H|} = 1 - \frac{T_C}{T_H}, \qquad K_C = \frac{T_C}{T_H - T_C}$$

T in **kelvin**. The refrigerator form follows from $K = |Q_C|/(|Q_H| - |Q_C|)$ with $|Q_C|/|Q_H| = T_C/T_H$.

Physical meaning

e_C is large when the temperature difference is large and very small when the temperatures are nearly equal; it equals 1 only for $T_C = 0$.

Common error

Using $^\circ\mathrm{C}$. All Carnot calculations need kelvin, because they come from $pV = nRT$.

But what if... you want each symbol and its limits checked before using them? Read both results in the **equation anatomy**.

EQUATION – Carnot engine

$$e_C = 1 - \frac{T_C}{T_H}$$

Meaning. the largest possible thermal efficiency between two reservoirs: that of a Carnot engine.

Variables. T_H, T_C in kelvin.

Units. none.

Interpretation. rises as T_C/T_H falls; below 1 for any finite $T_C > 0$; equals $1 - |Q_C|/|Q_H|$ only for a reversible engine.

Example. $600\,\mathrm{K}$ and $300\,\mathrm{K}$: 0.50.

EQUATION – Carnot refrigerator

$$K_C = \frac{T_C}{T_H - T_C}$$

Meaning. the largest coefficient of performance between two temperatures.

Variables. T_H, T_C in kelvin.

Units. none.

Interpretation. large when the temperature difference is small.

Example. freezer, $255\,\mathrm{K}$ and $298\,\mathrm{K}$: 5.93.

But what if... limits must be applied with a method? A procedure: **Toolkit 7**.

- **TOOLKIT 7 – The Carnot limit**

1. IDENTIFY the two reservoirs and convert both to kelvin.
2. LIMIT: $e_{\max} = 1 - T_C/T_H$; for a refrigerator $K_{\max} = T_C/(T_H - T_C)$.
3. TEST a claim: compare the claimed e (or K) with the limit; or compute $\Delta S_{\text{total}} = |Q_C|/T_C - |Q_H|/T_H$ per cycle.
4. DECIDE: $e > e_C$ or $\Delta S_{\text{total}} < 0$: impossible. $e = e_C$: reversible. $e < e_C$: allowed, irreversible.
WATCH OUT: °C in the formula; treating e_C as a design target (reversible means infinitely slow, with zero power).
CHECK: $0 < e < e_C < 1$.

But what if... the test uses both e_C and ΔS_{total}, and they always agree? Why? **Entropy and the Carnot limit**.

Entropy and the Carnot limit

Per cycle the engine returns to its start ($\Delta S_{\text{engine}} = 0$). The reservoirs change by

$$\Delta S_{\text{total}} = \frac{|Q_C|}{T_C} - \frac{|Q_H|}{T_H}.$$

Reversible ($|Q_C|/|Q_H| = T_C/T_H$): $\Delta S_{\text{total}} = 0$.
Irreversible: $\Delta S_{\text{total}} > 0$, so

$$\frac{|Q_C|}{|Q_H|} > \frac{T_C}{T_H} \;\Rightarrow\; e < 1 - \frac{T_C}{T_H}.$$

Physical meaning

The second law in terms of entropy gives the efficiency limit directly. Any irreversibility generates entropy and lowers the work obtained.

One line

$e \le e_C$ and $\Delta S_{\text{total}} \ge 0$ say the same thing.

But what if... irreversibility generates entropy and lowers the work? Then how much work is lost? **Entropy generated and lost work**.

Rule: entropy generated and lost work

Per cycle: $\Delta S_{\text{gen}} = \Delta S_{\text{total}} = \dfrac{|Q_C|}{T_C} - \dfrac{|Q_H|}{T_H} \geq 0$, and
$W_{\text{lost}} = e_C|Q_H| - W = T_C\,\Delta S_{\text{gen}}$.
$e_C|Q_H|$ is the work a Carnot engine would deliver from the same $|Q_H|$. For a plant, use rates: $\dot{S}_{\text{gen}} = P_C/T_C - P_H/T_H$ and $P_{\text{lost}} = T_C\dot{S}_{\text{gen}}$.

Why: $T_C\Delta S_{\text{gen}} = |Q_C| - |Q_H|\dfrac{T_C}{T_H} = (|Q_H| - W) - (1 - e_C)|Q_H| = e_C|Q_H| - W.$

Example: an engine absorbs $2000\,\text{J}$ at $600\,\text{K}$ and rejects $1200\,\text{J}$ at $300\,\text{K}$:
$\Delta S_{\text{gen}} = \dfrac{1200}{300} - \dfrac{2000}{600} = +0.67\,\text{J/K}$, so $W_{\text{lost}} = (300)(0.67) = 200\,\text{J}$. Check:
$e_C|Q_H| - W = (0.50)(2000) - 800 = 200\,\text{J}$. ✓

But what if... both tests now give the same verdict on a real claim? Apply them together to an inventor's engine: **Example 8**.

Question

An inventor claims an engine operating between reservoirs at $600\,\mathrm{K}$ and $300\,\mathrm{K}$ that absorbs $4000\,\mathrm{J}$ and delivers $2300\,\mathrm{J}$ of work per cycle. Is this possible?

Predict first

Sketch the arrows. Is $e = 2300/4000$ more or less than $\frac{1}{2}$? What would be the largest possible e for these reservoirs?

Limit: $e_C = 1 - 300/600 = 0.50$. **Claim:** $e = 2300/4000 = 0.575 > 0.50$.

Check by entropy: $|Q_C| = 4000 - 2300 = 1700\,\mathrm{J}$;

$$\Delta S_{\text{total}} = \frac{1700}{300} - \frac{4000}{600} = 5.67 - 6.67 = -1.0\,\mathrm{J/K} < 0.$$

Decision: **impossible**: it exceeds the Carnot limit and lowers the total entropy. At most $2000\,\mathrm{J}$ of work could be obtained.

But what if... limits matter in practice too? Compare real devices with their limits: **Example 9**.

Power station: steam at $500\,^\circ\mathrm{C}$, river at $20\,^\circ\mathrm{C}$. $T_H = 773.15\,\mathrm{K}$, $T_C = 293.15\,\mathrm{K}$:

$$e_C = 1 - \frac{293.15}{773.15} = 0.621.$$

Real stations reach about 0.35 to 0.45: friction, finite temperature differences and heat losses make the cycle irreversible.

Freezer: interior $-18\,^\circ\mathrm{C}$, kitchen $25\,^\circ\mathrm{C}$. $T_C = 255.15\,\mathrm{K}$, $T_H = 298.15\,\mathrm{K}$:

$$K_C = \frac{255.15}{43.0} = 5.93.$$

To remove $200\,\mathrm{W}$ of heat the work input is at least $200/5.93 = 34\,\mathrm{W}$. Real freezers have $K \approx 2$.

Predict first

Both limits are upper bounds. Is each actual device above or below its limit?

But what if... an engineer can change only one temperature? A **concept check** asks which change helps most.

Choose, then explain

A Carnot engine works between $T_H = 600\,\mathrm{K}$ and $T_C = 300\,\mathrm{K}$. Which change raises e more? (i) Raise T_H by $50\,\mathrm{K}$. (ii) Lower T_C by $50\,\mathrm{K}$.

A. (i) B. (ii) C. both equal D. neither changes e

Predict first

Write $e = 1 - T_C/T_H$. Which variable changes the ratio more for the same $50\,\mathrm{K}$? Estimate before calculating.

Base: $e = 0.500$. (i) $1 - 300/650 = 0.538$. (ii) $1 - 250/600 = 0.583$. **B**: lowering T_C by ΔT changes the ratio by $\Delta T/T_H$, while raising T_H changes it only by $T_C \Delta T/T_H^2$, which is smaller when $T_C < T_H$.

But what if... you can now judge limits unaided? Try an independent problem: **your turn**.

Independent problem

An engine absorbs $1500\,\mathrm{J}$ from a reservoir at $500\,\mathrm{K}$ and rejects $1000\,\mathrm{J}$ to a reservoir at $300\,\mathrm{K}$ in each cycle. Decide whether it is possible, whether it is reversible, and find ΔS_{total} per cycle.

Predict first

Sketch the arrows. Compare e with $e_C = 1 - 300/500$ before the entropy check.

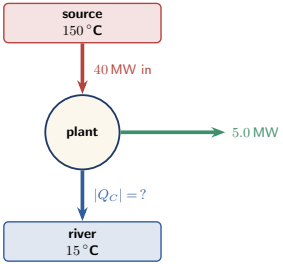
$W = 1500 - 1000 = 500\,\mathrm{J}$, $e = 0.333 < e_C = 0.400$.

$$\Delta S_{\text{total}} = \frac{1000}{300} - \frac{1500}{500} = 3.33 - 3.00 = +0.33\,\mathrm{J/K} > 0.$$

Evaluate: allowed but irreversible, since $e < e_C$ and $\Delta S_{\text{total}} > 0$ agree. ✓

But what if... real problems do not say which law to use? An **integrated problem** mixes them: Example 10.

Example 10: an integrated problem



Application problem

A geothermal plant takes heat from hot water at $150^\circ\mathrm{C}$ and rejects it to a river at $15^\circ\mathrm{C}$. It receives heat at $40\,\mathrm{MW}$ and delivers $5.0\,\mathrm{MW}$ of electrical power.

(a) What is the maximum power any engine could deliver from this heat?
(b) Find the plant's efficiency and the rate at which it rejects heat.
(c) Find the rate at which entropy is generated.
(d) How much of the available power is lost to irreversibility?

Predict first

Sketch the energy flow. Is the actual e above or below the maximum? Is the lost power more or less than half the maximum?

Example 10 restated

A geothermal plant takes heat from hot water at $150^\circ\mathrm{C}$ and rejects it to a river at $15^\circ\mathrm{C}$. It receives heat at $40\,\mathrm{MW}$ and delivers $5.0\,\mathrm{MW}$ of electrical power. (a) What is the maximum power any engine could deliver from this heat? (b) Find the plant's efficiency and the rate at which it rejects heat. (c) Find the rate at which entropy is generated. (d) How much of the available power is lost to irreversibility?

Identify, set up: two reservoirs (Toolkit 7), energy flow (Toolkit 4) and entropy (Toolkit 3). $T_H = 423.15\,\mathrm{K}$, $T_C = 288.15\,\mathrm{K}$.

(a) $e_C = 1 - \dfrac{288.15}{423.15} = 0.319; \quad P_{\max} = e_C P_H = (0.319)(40) = 12.8\,\mathrm{MW}.$

(b) $e = \dfrac{5.0}{40} = 0.125; \quad P_C = 40 - 5.0 = 35\,\mathrm{MW}.$

But what if... the plant delivers far less than the maximum, and parts (c) and (d) ask where the difference went? Use entropy generated and lost work.

Asked: (c) the rate at which entropy is generated; (d) how much of the available power is lost to irreversibility. *From part (1):* $P_H = 40\,\mathrm{MW}$ at $T_H = 423.15\,\mathrm{K}$, $P_C = 35\,\mathrm{MW}$ at $T_C = 288.15\,\mathrm{K}$, $P = 5.0\,\mathrm{MW}$, $P_{\max} = 12.8\,\mathrm{MW}$.

(c) $\dot{S}_{\text{gen}} = \dfrac{P_C}{T_C} - \dfrac{P_H}{T_H} = \dfrac{35\times10^6}{288.15} - \dfrac{40\times10^6}{423.15} = 1.215\times10^5 - 0.945\times10^5 = 2.7\times10^4\,\mathrm{W/K}.$

(d) $T_C\dot{S}_{\text{gen}} = (288.15)(2.69\times10^4) = 7.8\,\mathrm{MW}.$

Evaluate: $e = 0.125 < e_C = 0.319$ (allowed); $\dot{S}_{\text{gen}} > 0$; and $P_{\max} - P = 12.8 - 5.0 = 7.8\,\mathrm{MW}$, equal to $T_C\dot{S}_{\text{gen}}$: the lost work is T_C times the entropy generated. About 61% of the available power is lost. ✓

But what if... the same limit applies when the device is run backward to heat a house? An extension: the **heat pump**.

Extension (unassessed): the heat pump

A heat pump is a refrigerator whose *hot* side is the house. Its performance is

$$K_{\text{hp}} = \frac{|Q_H|}{W} \le \frac{T_H}{T_H - T_C}.$$

For a house at 22°C and outside air at 2°C:

$$K_{\text{hp,max}} = \frac{295.15}{20.0} = 14.8.$$

Why it beats a heater

An electric heater turns $1\,\text{J}$ of work into $1\,\text{J}$ of heat. A heat pump with $K_{\text{hp}} = 3$ delivers $3\,\text{J}$ of heat for $1\,\text{J}$ of work, taking the rest from the outside air.

An application of Toolkits 6 and 7; it is not assessed. Real heat pumps reach about 3 to 4.

But what if... where does the second law lead beyond this course? **Connect** the ideas.

The arrow of time

Entropy is the quantity that distinguishes past from future in an isolated system; $S = k \ln w$ links it to counting arrangements (supplementary notes).

Technology

Every power station, engine, refrigerator and heat pump is limited by e_C and K_C; this sets the target for efficiency improvements.

Further physics

Statistical mechanics, chemical thermodynamics and the physics of information build on the entropy rule met here.

Physical meaning

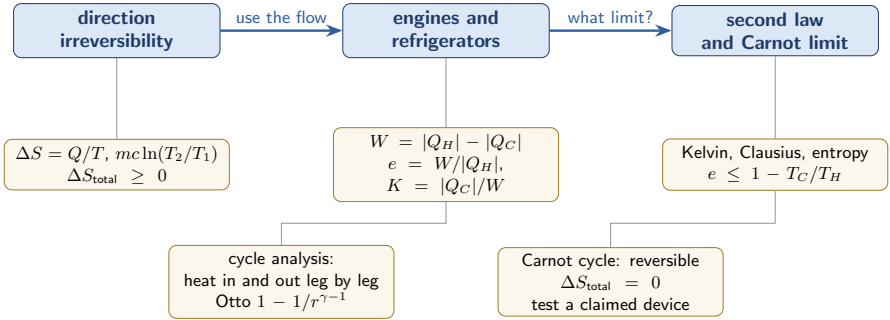
Together with energy conservation, the second law completes the thermal half of the course: energy is conserved, but its *usefulness* is not.

But what if... ideas used well still fail at a few predictable points? The last slide of Contact B collects them: **common mistakes**.

Common mistakes and the correct habit

Mistake	Instead
Thinking K cannot exceed 1.	$K = \lvert Q_C \rvert / W$ can exceed 1.
Thinking an open refrigerator cools the room.	The room warms by the work input.
Treating e_C as a design target.	It is an upper bound that needs infinite time.
Using $^\circ\mathrm{C}$ in the Carnot formula.	Kelvin.
Believing a better working substance raises the limit.	Every reversible engine has the same e_C.
Testing a claim with energy conservation alone.	Compare with e_C and ΔS_{total}.
Forgetting that real engines are irreversible.	$e < e_C$ and $\Delta S_{\text{total}} > 0$.

But what if... the habits are in place? Pull the whole lecture together in the summary.


direction irreversibility
use the flow
engines and refrigerators
what limit?
second law and Carnot limit
$\Delta S = Q/T$, $mc\ln(T_2/T_1)$
$\Delta S_{\text{total}} \geq 0$
$W = |Q_H| - |Q_C|$
$e = W/|Q_H|$,
$K = |Q_C|/W$
Kelvin, Clausius, entropy
$e \leq 1 - T_C/T_H$
cycle analysis:
heat in and out leg by leg
Otto $1 - 1/r^{\gamma-1}$
Carnot cycle: reversible
$\Delta S_{\text{total}} = 0$
test a claimed device

The question we started with

Why do natural processes run in one direction only, and what limits how much of the heat from a hot source an engine can turn into work?

Direction (Contact A)

Every natural process is irreversible. When all systems that take part are included, the total entropy never decreases ($\Delta S_{\text{total}} \geq 0$), so processes run only in the direction that raises it.

Conversion and the limit (Contacts A and B)

An engine must reject heat, so $e < 1$; no engine does better than a Carnot engine, $e \leq e_C = 1 - T_C/T_H$ with T in kelvin.

Direction $\Rightarrow$ **Conversion** (engines and refrigerators, $W = |Q_H| - |Q_C|$) $\Rightarrow$ **Limit** (second law, Carnot).

MASTER PROCEDURE – SECOND LAW

1. IDENTIFY The system, the reservoirs and the process: heat flow, engine, refrigerator or expansion.
2. SKETCH The energy-flow diagram or the pV cycle, with the arrows.
3. UNITS AND SIGNS Kelvin; magnitudes of Q_H and Q_C; signed Q in a ledger.
4. PREDICT Allowed or not; the sign of ΔS; bounds on e or K.
5. SELECT THE MODEL Entropy rule, energy flow, cycle ledger, Otto or Carnot limit.
6. EXECUTE Write the equations first, then substitute with units.
7. EVALUATE Units, $W = |Q_H| - |Q_C|$, $0 < e < e_C$, and $\Delta S_{\text{total}} \geq 0$.
SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

The thermal half of the course is now complete: temperature, gases and phases, energy conservation, and the direction and limits of energy conversion.

THE SEVEN TOOLKITS

1	**Reversible or irreversible?**	the film test; causes; first law gives no direction
2	**Entropy change**	$\Delta S = Q/T$; $\pm mL/T$; $mc\ln(T_2/T_1)$; kelvin
3	**Entropy bookkeeping**	system, surroundings, total; $\Delta S_{\text{total}} \geq 0$
4	**Energy flow in an engine**	$W = \lvert Q_H\rvert - \lvert Q_C\rvert$; $e = W/\lvert Q_H\rvert$; magnitudes
5	**Cycle analysis for efficiency**	ledger; sort legs into heat in and heat out; Otto
6	**Refrigerators and heat pumps**	$\lvert Q_H\rvert = \lvert Q_C\rvert + W$; $K = \lvert Q_C\rvert/W$
7	**The Carnot limit**	$e_C = 1 - T_C/T_H$; K_C; test a claim

The Master Procedure sits above these. Earlier toolkits (units, signs, the ledger, adiabatic scaling, $Q = mc\Delta T$) are reused, not renamed.

Summary: key ideas

Key ideas

- The first law gives no direction; the second law does.
- The entropy of a part can fall; the total cannot.
- Every cyclic engine rejects heat: $e < 1$.
- $e \leq e_C = 1 - T_C/T_H$; $K \leq K_C$.
- Entropy generated is lost work divided by T_C.
- Use kelvin and magnitudes.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact (Contact A has one for direction and one for engines).

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Entropy, isothermal	$\Delta S = Q/T$	reversible; J/K
Entropy, heating	$\Delta S = mc\ln(T_2/T_1)$	kelvin
Second law	$\Delta S_{\text{total}} \geq 0$	system plus surroundings
Heat engine	$W = \lvert Q_H\rvert - \lvert Q_C\rvert, \quad e = W/\lvert Q_H\rvert$	magnitudes
Otto cycle	$e = 1 - 1/r^{\gamma-1}$	ideal gas
Refrigerator	$\lvert Q_H\rvert = \lvert Q_C\rvert + W, \quad K = \lvert Q_C\rvert/W$	K can exceed 1
Carnot	$e_C = 1 - T_C/T_H, \quad K_C = T_C/(T_H - T_C)$	kelvin
Generated entropy	$\Delta S_{\text{gen}} = \lvert Q_C\rvert/T_C - \lvert Q_H\rvert/T_H$	lost work $= T_C \Delta S_{\text{gen}}$

Checklist: can I . . .

CORE

- ☐ distinguish reversible from irreversible and name the causes?
- ☐ calculate $\Delta S = Q/T$ and $\Delta S = mc\ln(T_2/T_1)$?
- ☐ find ΔS of system, surroundings and total?
- ☐ decide from ΔS_{total} whether a process is allowed?
- ☐ draw an energy-flow diagram and use $W = |Q_H| - |Q_C|$?
- ☐ calculate $e = W/|Q_H|$ and engine power?
- ☐ find the heat in and out on each leg of a pV cycle?
- ☐ derive and use the Otto efficiency?
- ☐ use $K = |Q_C|/W$ for a refrigerator and a heat pump?
- ☐ state the three forms of the second law?
- ☐ describe the four stages of the Carnot cycle?
- ☐ use $e_C = 1 - T_C/T_H$ and K_C?
- ☐ test a claimed engine with e_C and ΔS_{total}?
- ☐ select the right physics in an unfamiliar problem?

After this week

Tutorial: entropy, mixing, engine, Otto, refrigerator, Carnot and combined problems. Supplement: equivalence of the statements, entropy of an ideal gas, $S = k\ln w$, the Diesel cycle, the heat pump, the arrow of time.

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